Circles as Conic Sections

Algebra & Geometry Worksheet · Grade 9–11

Name: ________________________________ Date: ___________________

Learning Objectives

- Identify the center and radius of a circle from its standard equation
- Write the standard equation of a circle given its center and a point on the circle
- Convert the general form of a circle to standard form using completing the square

Problems

1. Identify the center and radius of the circle given by the equation below.

$$(x-3)^2+(y-5)^2=49$$

2. Identify the center and radius of the circle given by the equation below.

$$(x+4)^2+(y-1)^2=25$$

3. Write the standard equation of a circle with the center and radius given below.

$$\text{Center: } (2,\,-6), \quad r=3$$

4. Describe what value of r represents in the standard equation of a circle, then find the radius of the circle given below.

$$(x+2)^2+(y-7)^2=36$$

5. A circle has its center at the origin and passes through the point given below. Write the standard equation of the circle.

$$\text{Point on circle: } (0,5)$$

6. Find the equation of the circle whose center and a point on the circle are given below.

$$\text{Center: } (-3, 4), \quad \text{Point: } (1, 4)$$

7. A circle has center at the point below and passes through the point shown. Find the standard equation of the circle.

$$\text{Center: } (-5, 0), \quad \text{Point: } (-6, 4)$$

8. Convert the general equation of the circle below to standard form, then state the center and radius.

$$x^2 + y^2 - 6x + 4y - 3 = 0$$

9. Convert the general equation of the circle below to standard form, then state the center and radius.

$$x^2 + y^2 - 4x + 8y - 5 = 0$$

10. Convert the general equation below to standard form, identify the center and radius, then list the four key points used to graph the circle.

$$x^2 + y^2 + 10x - 2y + 1 = 0$$

Circles as Conic Sections — Answer Key

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Answer Key

1. Answer: Center: (3, 5), Radius: 7

- Compare with $(x - h)^2 + (y - k)^2 = r^2$
- $h = 3$, $k = 5$, so center = (3, 5)
- $r^2 = 49$, so $r = 7$

2. Answer: Center: (-4, 1), Radius: 5

- Rewrite as $(x - (-4))^2 + (y - 1)^2 = 25$
- $h = -4$, $k = 1$, so center = (-4, 1)
- $r^2 = 25$, so $r = 5$

3. Answer: $(x - 2)^2 + (y + 6)^2 = 9$

- Use the standard form $(x - h)^2 + (y - k)^2 = r^2$
- Substitute $h = 2$, $k = -6$, $r = 3$
- $(x - 2)^2 + (y + 6)^2 = 9$

4. Answer: r is the distance from the center to any point on the circle; $r = 6$

- r represents the fixed distance from the center (h, k) to any point on the circle
- $r^2 = 36$
- $r = 6$

5. Answer: $x^2 + y^2 = 25$

- Center is (0, 0), so $h = 0$, $k = 0$
- Substitute the point (0, 5) into $(x - 0)^2 + (y - 0)^2 = r^2$
- $0 + 25 = r^2$, so $r^2 = 25$
- Equation: $x^2 + y^2 = 25$

6. Answer: $(x + 3)^2 + (y - 4)^2 = 16$

- Find r using the distance formula: $r = \sqrt{(1-(-3))^2 + (4-4)^2}$
- $r = \sqrt{16 + 0} = 4$, so $r^2 = 16$
- Substitute $h = -3$, $k = 4$: $(x + 3)^2 + (y - 4)^2 = 16$

7. Answer: $(x + 5)^2 + y^2 = 17$

- Use $(x - h)^2 + (y - k)^2 = r^2$ with $h = -5$, $k = 0$
- Substitute the point (-6, 4): $(-6 + 5)^2 + (4 - 0)^2 = r^2$
- $(-1)^2 + 4^2 = 1 + 16 = 17 = r^2$
- Equation: $(x + 5)^2 + y^2 = 17$

8. Answer: $(x - 3)^2 + (y + 2)^2 = 16$; Center: (3, -2), Radius: 4

- Group and move constant: $(x^2 - 6x) + (y^2 + 4y) = 3$

- Complete the square for x: take half of -6 = -3, square it = 9; add 9 to both sides
- Complete the square for y: take half of 4 = 2, square it = 4; add 4 to both sides
- $(x - 3)^2 + (y + 2)^2 = 3 + 9 + 4 = 16$
- Center: (3, -2), r = 4

9. Answer: $(x - 2)^2 + (y + 4)^2 = 25$; Center: (2, -4), Radius: 5

- Group and move constant: $(x^2 - 4x) + (y^2 + 8y) = 5$
- Complete the square for x: $(\text{half of } -4)^2 = 4$; add 4 to both sides
- Complete the square for y: $(\text{half of } 8)^2 = 16$; add 16 to both sides
- $(x - 2)^2 + (y + 4)^2 = 5 + 4 + 16 = 25$
- Center: (2, -4), r = 5

10. Answer: $(x + 5)^2 + (y - 1)^2 = 25$; Center: (-5, 1), Radius: 5; Key points: (-10, 1), (0, 1), (-5, 6), (-5, -4)

- Group and move constant: $(x^2 + 10x) + (y^2 - 2y) = -1$
- Complete the square for x: $(\text{half of } 10)^2 = 25$; add 25 to both sides
- Complete the square for y: $(\text{half of } -2)^2 = 1$; add 1 to both sides
- $(x + 5)^2 + (y - 1)^2 = -1 + 25 + 1 = 25$
- Center: (-5, 1), r = 5
- Key points: left (-10,1), right (0,1), up (-5,6), down (-5,-4)